

semiconductor (CMOS) image sensors of the mobile camera do not sense red, green, and blue wavelengths for each pixel. It only senses one colour for each. Now, it is the job of image signal processors (ISPs) to interpolate this image by applying methods of computing. In addition, the ISP does all other housekeeping; it controls autofocus, exposure, and white balance for the camera system. Moreover, modern image processors can correct lens imperfections as well as things like HDR recombining, noise reduction, other filtering, face or object detection, and conversion between colour spaces.

The latest Intel i9 processors integrate an integrated ISP with advanced hardware implementation that improves image and video quality of cameras. The Apple M1 features Apple's latest ISP for higher-quality video with better noise reduction, greater dynamic range, and improved auto white balance.

Advanced security features: The significance of security features is that they deal with today's most sophisticated attacks and viruses. The memory protection systems together provide real-time encryption of system memory for protection against physical attacks.

Latest laptop processors include high-performance AES (advanced encryption standards) encryption hardware that enables fast and secure data encryption and decryption. The hardware based encryption is done to and from the integral encrypted solid-state drive (SSD) and it is much more complex than similar software encryption methods. Higher complexity means better security; hardware based encryption is much more secure and faster as compared to software based encryption systems. The AES 256 can perform encryptions at the rate of 10Gbps without significant latency.

Intel's AES New-Instructions (AES-NI) is featured in its latest processors for a wide range of cryptographic applications such as applications that perform bulk encryption/decryption, authentication, random number generation, and authenticated encryption.

Apple M1's high-performance storage controller includes AES encryption hardware for faster and more secure SSD performance.

Apple M1 processors integrate the latest Secure Enclave for enhanced security. Secure Enclave is a hardware component, more like a coprocessor, integrated in almost all Apple chips. It runs a microkernel and boots it separately from the rest of the device. During this boot sequence, the Secure Enclave generates an encryption key valid for a very short time and embeds it with a user ID that is inaccessible by rest of the processor. The key is used to verify authenticity of any data written in the device's memory by the Secure Enclave. When authentication data is sent from biometric sensors to Secure Enclave, it authenticates the data and sends the CPU the notification. The CPU then permits the user to unlock or make purchases with the device.

Intel processors include many more security features such as:

- **Intel software guard extension (Intel SGX):** It allows applications to create hardware-enforced trusted execution protection for their sensitive routines and data. It allows the developers to partition sensitive information into enclaves (the areas of execution in memory with more security protection). The enclaves have hardware-assisted confidentiality and integrity-added protections that prevent access to this sensitive data.
- **Intel boot guard:** It helps protect the system's pre-OS environment from viruses and malicious software attacks.
- **Intel control-flow enforcement technology (CET):** It helps prevent misuse and exploits from hijacking the control-flow transfer instructions.
- **Intel OS guard:** It does not allow

the operating system (OS) to directly execute the application code in order to reduce chances of attacks on the OS.

- **Mode based execute control (MBE):** It authenticates kernel-level code.

The Snapdragon 8cx Gen2 processor's integrated security protects passwords and log-in data with full Windows 10 support. It supports facial and fingerprint ID. The cellular 4G and 5G connectivity eliminates the need for relying on unsecure public Wi-Fi networks while allowing untethered, remote connection.

5G connectivity

Qualcomm processors enable 4G and 5G connectivity in laptops. They integrate the Snapdragon X55 5G Modem-RF system, supporting 5G (sub-6 GHz and mmWave), 4G LTE, and Wi-Fi 6. The cellular connectivity in laptops eliminates the need for trusted Wi-Fi networks. Whenever needed, the system automatically switches between the trusted Wi-Fi and cellular networks, depending on location. The wireless modem-RF system also receives accurate navigation and location based services with location awareness.

Conclusion

The latest processors are improving every aspect of performance and efficiency. As much as they are getting faster, they are also more energy-efficient and reliable. They are now capable of machine learning and deep learning, ensuring reliability and efficient operation—regardless of the workload. Also, the processors are integrating more high-performance GPUs for gaming and video streamings. In a nutshell, they are becoming way better in terms of speed, performance, and power consumption. **END** 🐼

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